QSME-Net: Quad-State Mamba-Enhanced Network for Hyperspectral Anomaly Detection

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***Abstract*—To overcome the limitations of current hyperspectral anomaly detection (HAD) methods, where imperfect matching mechanisms hinder the effective separation of anomalies from the background, this article presents an innovative network named quad-state Mamba-enhanced network (QSME-Net). The core of this approach is to improve the precision of spectral matching and the efficacy of background reconstruction via joint spatial–spectral frequency analysis. It features two innovative modules: the spectral-domain matcher (SDM) and the quad-state cross enhancer (QSCE), integrated with the state-space models (SSMs). SDM utilizes the Fourier transform to extract the spatial frequency components from local patches and combines them with spectral signatures to compute block similarities, ensuring accurate spectral matching. QSCE adaptively fuses multiscale and global–local features, guided by spectral characteristics, to improve the anomaly and background representation. By incorporating Mamba's selective scanning and Transformer-based attention, QSME-Net dynamically balances local and global feature interactions, optimizing contextual modeling for robust background reconstruction. The empirical evidence from experiments on six real hyperspectral images (HSIs) substantiates the superior detection efficacy of our proposed methodology in contrast to several contemporary, high-performing HAD algorithms. The source code of QSME-Net is available at: https://github.com/IIP-Team**

***Index Terms*—Hyperspectral anomaly detection (HAD), quad-state cross enhancer (QSCE), spectral-domain matcher (SDM), state-space models (SSMs).**

I. Introduction

In contrast to traditional remote sensing, the hyperspectral imaging (HSI) simultaneously acquires radiative, geometric, and spectral information of surface objects [1], [2]. Its high-resolution spectral data allows for the differentiation of subtle spectral variations in ground targets [3], which has spurred the widespread application of hyperspectral technology [4], [5]. Within the core HSI research field of target detection, methods can be generally categorized into spectral matching and anomaly detection. The spectral matching relies on the prior information of target spectral signatures for identification [6]. In contrast, the anomaly detection does not require prior information, identifying targets solely based on their spectral differences from the background [7]. Given the practical difficulty of acquiring prior spectral data, anomaly detection offers greater utility and has evolved into a foundational HSI processing task.

In HSIs, an "anomaly" is typically defined as a small target whose spectral features differ significantly from its neighboring pixels [8]. A fundamental detection approach involves calculating the background mean and flagging the most distant pixels as anomalies. This forms the basis of statistical methods, which depend on stringent assumptions about the background data's distribution [9]. However, the complexity and nonlinear nature of HSI data frequently render these assumptions invalid. To overcome this limitation, some techniques focus on constructing statistical models that better align with the actual data distribution, while others leverage machine learning to perform the anomaly detection without relying on specific background distribution hypotheses [10].

Deep learning-based methods dominate hyperspectral anomaly detection (HAD) due to strong feature modeling, avoiding reliance on prior distribution assumptions or manual parameter tuning. They use self-supervised HSI reconstruction: the background is accurately reconstructed, and anomalies' larger reconstruction errors serve as detection indicators. However, the assumption that anomalies have much larger errors than the background is not always valid—models prioritize simple data distributions and struggle with complex backgrounds. Feature constraints can solve this: enhancing background features to improve the reconstruction and weakening anomaly features to avoid overfitting, thus boosting detection reliability and accuracy.

Several recent deep learning techniques guide HSI reconstruction by incorporating specific constraints. For example, the Auto-AD network [10] is predicated on using reconstruction errors as adaptive weights to inhibit the generation of anomalies. A significant drawback of this method is its struggle with complex backgrounds, where reconstruction inaccuracies can inadvertently degrade the background representation itself. Similarly, the DeepLR method [11] presumes a low-rank (LR) structure for the reconstructed back-

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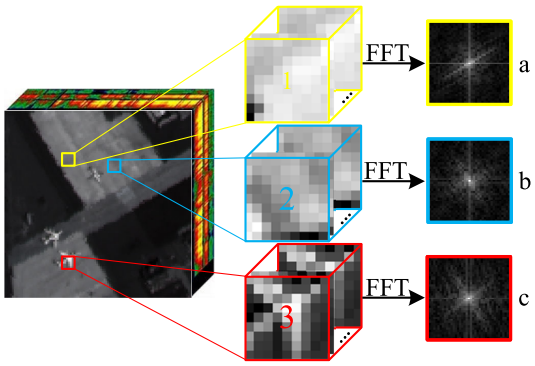


Fig. 1. Background blocks 1 and 2 and anomaly block 3. Photographs a–c are their frequency spectra.

ground. However, imposing this LR constraint often erases high-frequency details, which in turn leads to a higher rate of false alarms (FARs). The principal weakness shared by these methods is an overemphasis on suppressing anomalies at the expense of high-fidelity background reconstruction. This imbalance restricts their effectiveness and underscores the need for more robust HAD models.

These limitations, stemming from what we define as "imperfect matching mechanisms," can be broadly categorized. The first and most critical issue for reconstruction-based methods is anomaly contamination, or "over-matching." As these networks (such as Auto-AD) are optimized to minimize reconstruction error across all pixels, they inadvertently learn the latent characteristics of the anomaly targets. This "over-matching" phenomenon allows the network to reconstruct true anomalies with high fidelity, diminishing the residual error and leading to a high rate of missed detections.

The second limitation is poor background modeling or "under-matching." This occurs when a model imposes overly simplistic constraints, such as the LR structure in DeepLR, or fails to represent complex heterogeneous areas. Such models inherently fail to accurately capture the true background distribution. Consequently, background pixels exhibiting unique spectral signatures that do not conform to the oversimplified model are frequently misclassified as anomalies, resulting in a high FAR.

Therefore, an ideal HAD constraint mechanism must simultaneously solve both problems: it must be precise enough to fully model the complex background (preventing "under-matching") while being discriminative enough to suppress the learning of anomaly features (avoiding "over-matching").

The foundational premise of HAD is the pronounced spectral contrast between anomalies and the background. Anomalies are defined by their spectral dissimilarity to their immediate surroundings, whereas background regions are characterized by a high degree of spectral coherence. This inherent dichotomy allows content similarity to be used as a feature constraint, guiding a model to enhance background features while suppressing anomaly features for an optimized reconstruction. However, relying solely on spectral similarity struggles to distinguish dynamic backgrounds from subtle anomalies in complex scenes. Fig. 1 presents the spatial frequency analysis of three representative blocks from the SanDiego hyperspectral dataset, where blocks 1 and 2 represent background regions, and block 3 denotes an anomalous region, accompanied by their respective spatial frequency spectra (a–c). Obtained via the fast Fourier transform (FFT) [12], these spectra reveal that background blocks are dominated by low-frequency components (high central brightness in a and b, with rapid decay toward the edges), whereas the anomalous block exhibits pronounced high-frequency components (slower brightness decay in c, with significant edge brightness). This disparity underscores the pivotal role of capturing spatial–spectral structures in the hyperspectral analysis. Consequently, we opt to incorporate usage frequency information, leveraging the frequency distribution patterns of background and anomalies across spatial and temporal dimensions to model background behavior more precisely and enhance anomaly detection accuracy.

In this article, we propose an innovative quad-state Mamba-enhanced network (QSME-Net) method for HAD, featuring two original core modules: Spectral-domain matcher (SDM) and quad-state cross enhancer (QSCE), integrated with the Mamba mechanism. The SDM employs the FFT to perform spatial frequency analysis on hyperspectral image (HSI) blocks. By transforming spatial patches into the frequency domain, it separates high-frequency anomaly details from low-frequency background information, while simultaneously preserving spectral consistency. QSCE combines multiscale and global–local feature extractors with anomaly and background enhancement modules (EMs), adaptively fusing anomaly and background information using spectral features to enhance the representation. Mamba dynamically integrates local and global features through selective scanning and Transformer-based attention, boosting contextual modeling. A thorough experimental evaluation reveals a significant performance advantage for our method when compared against current top-performing HAD techniques. The main contributions are summarized as follows.

1) We introduce QSME-Net, which employs the SDM to extract spectral signature frequency components via the Fourier transform and compute block similarities, enhancing HAD.
2) We develop an QSCE to adaptively fuse multiscale and global–local features, guided by spectral characteristics, optimizing anomaly and background representation.
3) We innovatively integrate the Mamba network's selective scanning mechanism with Transformer-based attention in QSME-Net, dynamically balancing local and global feature interactions to enhance contextual modeling for HAD.

This article is structured as follows. We begin by providing an overview of related work on HAD in Section II. Next, we introduce the details of our proposed QSME-Net in Section III. In Section IV, we present and analyze the results of our experiments. This article concludes with a summary of our work in Section V.

II. Related Works

A common taxonomy for HAD algorithms distinguishes between two primary approaches: those based on statistical

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models and those that are nonstatistical. Our subsequent review of the literature is organized around this distinction.

A. Statistics-Based HAD

Statistical approaches to HAD leverage the data's statistical attributes. The cornerstone of this category is the Reed–Xiaoli (RX) algorithm [13], which calculates the Mahalanobis distance from the background mean, assuming a Gaussian background distribution. However, RX is hampered by the inaccuracy of the Gaussian assumption for real scenes and by the contamination of background statistics by anomalies themselves. Improvements on RX have branched into two main directions: better background selection and background purification. The former includes methods like weighted RX [14], adaptive probability density estimation [15], and multiwindow analysis for capturing different anomaly sizes [16], [17]. The latter involves techniques to remove or mitigate anomaly influence, such as subspace modeling [18], iterative refinement [19], kurtosis-based testing [20], [21], and local linear embedding [22]. Beyond the RX-based family of algorithms, other statistical approaches have also been developed. For instance, the Gaussian Markov random field has been used to enhance the detection adaptability [23], while robust nonlinear detectors have been created by combining kernel learning [24] and robust regression analysis [25], [26]. Specific challenges have been targeted, such as subpixel detection via single-feature methods [27] and high-order spectral information extraction using the kernel principal component analysis (PCA) [28]. However, a fundamental limitation persists across most statistical approaches: their heavy reliance on underlying data assumptions. This dependency compromises their ability to generalize to new scenarios and makes them less effective in complex, noisy environments, where they may struggle to identify subtle anomalies, resulting in poor detection rates or high false alarms.

B. Nonstatistics-Based HAD

Approaches that do not rely on statistical models are generally grouped into three categories: distance-based, reconstruction-based, and deep-learning-based [10]. Within the distance-based category, the support vector data description (SVDD) is a foundational method [29]; it operates by fitting the smallest possible hypersphere around the data and using this boundary to distinguish anomalous points. Another set of distance-based techniques involves clustering, which has been applied to HAD in cases, where no prior information is available [30], [31]. Reconstruction-based techniques have recently become a prominent area of research in HAD [32], [33]. Many of these methods are based on the representation learning. For example, the collaborative representation-based detector (CRD) assumes background pixels can be represented by their local neighbors, while anomalies cannot [34]. Likewise, BJSRD uses a joint sparse representation to calculate anomaly scores from pixel weights [32]. Another major subcategory involves matrix decomposition. The LSMAD algorithm, for instance, models the background as an LR component and anomalies as a sparse component [35]. This concept was enhanced in the ADLR detector, which first applies spectral unmixing to create more distinct abundance vectors before performing a more accurate LR decomposition [36].

The use of deep learning in HAD has grown significantly in recent years [37]. These techniques can be grouped by their underlying concept. One group of methods uses deep networks as powerful feature extractors. This includes employing pretrained CNNs [38] or DBNs to capture high-order spectral characteristics [39]. Another major category is based on reconstruction. These methods often use autoencoders (AEs) to learn a compact representation of the background [40]. To enhance the reconstruction fidelity, some approaches integrate GANs with AEs to form adversarial AEs (AAEs) [41]. More complex, hybrid strategies have also emerged. The DSVDD classifier, for instance, transforms anomaly detection into a one-class classification problem by learning a hypersphere boundary in a deep feature space. Other hybrid methods use reconstruction networks to generate deep features, upon which a separate detection algorithm, such as an LR representation in the LREN model, is applied [42], [43]. Similarly, Siamese networks have been designed to learn transferable similarity metrics, moving beyond simple reconstruction [44]. Lian et al. [45] developed GT-HAD, leveraging the Transformer to exploit spatial–spectral similarity for the differential reconstruction of background and anomalies.

Recently, sophisticated deep reconstruction strategies have emerged to address the tendency of AEs to reconstruct anomalies. One significant branch employs self-supervised blind-spot networks, which reconstruct a pixel using only its surrounding context. To handle larger sized targets, PDB-SNet [46] introduces pixel-shuffle downsampling to break the anomaly's internal spatial correlation, while DirectNet [47] uses a "blind-block" inspired by the dual-window model, forcing reconstruction based only on the outer background context. PUNNet [48] further enhances this by injecting global contextual features via channel attention to improve background fidelity. Another distinct approach explores the transform domain. FTSGAN [49], a semi-supervised method, reframes HAD as a mapping task, training a GAN only on background samples to learn the inverse mapping from the fractional Fourier domain (FrFD). The network's inability to restore unseen anomaly patterns generates large, detectable errors.

A key limitation of the methods discussed above, despite their achievements, is the underutilization of the intrinsic properties of HSI data for distinguishing anomalies from the background. This oversight can restrict their effectiveness in certain scenarios. To overcome this, our proposed QSME-Net employs spectral-domain Fourier features and a self-supervised block-matching mechanism to achieve a more accurate reconstruction of high-spectral-resolution imagery, integrating multiscale feature fusion for robust data processing.

III. Methods

A. Overall Framework

QSME-Net is a deep learning framework designed for HSI anomaly detection. As illustrated in Fig. 2, the framework processes an input HSI denoted as $I \in \mathbb{R}^{H\times W\times C}$, where

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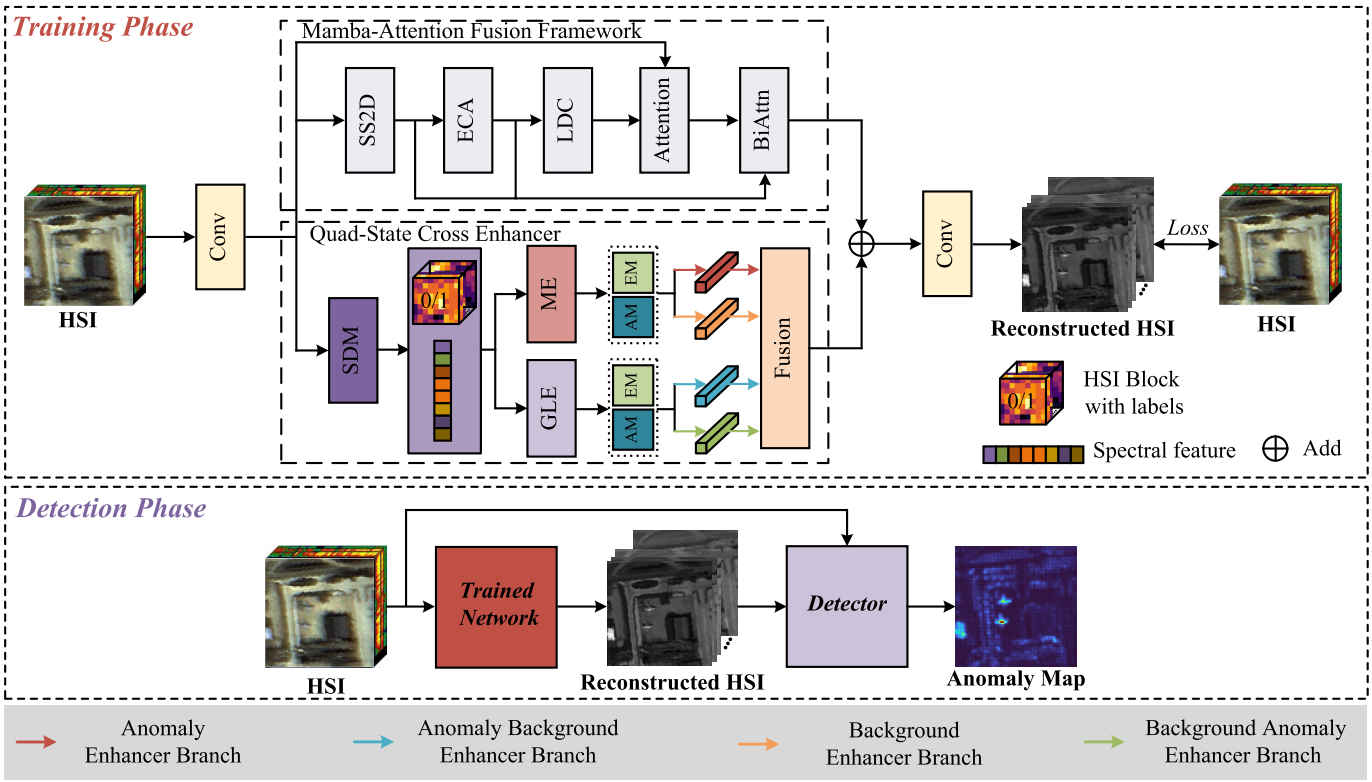


Fig. 2. Overall framework of the QSME-Net. During the training phase, the input HSI patches are processed by the SDM to generate initial matching labels via spectral frequency extraction. These labels and patches feed into the QSCE, where an ME and a global–local extractor (GLE) utilize Mamba's SS2D capture anomaly patterns and long-range background dependencies, respectively. Subsequently, the enhancement module (EM) and adaptation module (AM) generate four state features for robust discriminability. These features are fused and further refined by the MAFF—a cascade of SS2D and multiattention mechanisms—to achieve the high-fidelity HSI reconstruction optimized by L_2 loss. In the detection phase, the trained network reconstructs the test HSI, and a detector generates the final comprehensive anomaly map by calculating the reconstruction error.

H and W represent the spatial dimensions and C denotes the number of spectral bands. The architecture comprises three core components: the SDM, the QSCE, and a Mamba-enhanced encoder–decoder network.

To ensure the scalability and generalizability of the QSME-Net framework, we adopt a parameterized notation system for all spatial and spectral tensors. Let $k \times k$ be the spatial patch size, D be the feature channel dimension, and n be the dimensionality of the spectral feature vector extracted via the SFE unit. In the SDM, N represents the total number of patches in a scene used for background template ($\mathbf{F}_{template}$) initialization. Within the QSCE, the matched spectral features $\mathbf{F}_{x_j} \in \mathbb{R}^{B\times n}$ and labels $\mathbf{l} \in \{0,1\}^B$ guide the state generation. These parameterized variables maintain consistency across the dual-branch extraction and four state fusion stages, allowing the architecture to adapt to varying HSI datasets

1) *Patching and Preprocessing:*
To capture localized spatial–spectral characteristics, the input HSI is first decomposed into a set of overlapping patches $\{x_i\}$. Each patch $x_i \in \mathbb{R}^{C\times k\times k}$ contains all spectral bands for a spatial neighborhood of size $k \times k$. This decomposition is performed using a sliding window with a spatial kernel of size $k \times k$ and a stride s.
2) *Core Module Synergism:*
The framework transforms the input patches into an anomaly detection map through the following stages. (a,b,c)

a) *SDM:* The SDM employ s the FFT to perform joint spatial–spectral frequency analysis. It extracts signature frequency components, such as amplitude and energy, to generate initial matching labels $l_i \in \{0,1\}$, where 1 and 0 indicate the background and anomaly patches, respectively.
b) *QSCE:* Guided by the matching labels and spectral features from the SDM, the QSCE leverages Mamba's 2-D selective scanning (SS2D) module to generate and dynamically fuse four state features. This process optimizes the representation of both anomalies and the background.
c) *Mamba-Attention Fusion Framework (MAFF):* As the core of the encoder–decoder network, the MAFF refines these enhanced features to reconstruct the background patches. It integrates SS2D with Transformer-based attention to balance global contextual modeling and local feature interactions.

3) *Reconstruction and Anomaly Map Generation:*
During training, the SDM's background template and labels are iteratively updated based on the reconstructed patches $x_{out,i}$. For inference, the anomaly score S_i for each patch is calculated as the reconstruction error between the original patch x_i and its reconstruction $x_{out,i}$. Since the patches overlap, the final anomaly map $M \in \mathbb{R}^{H\times W}$ is generated by aggregating these scores

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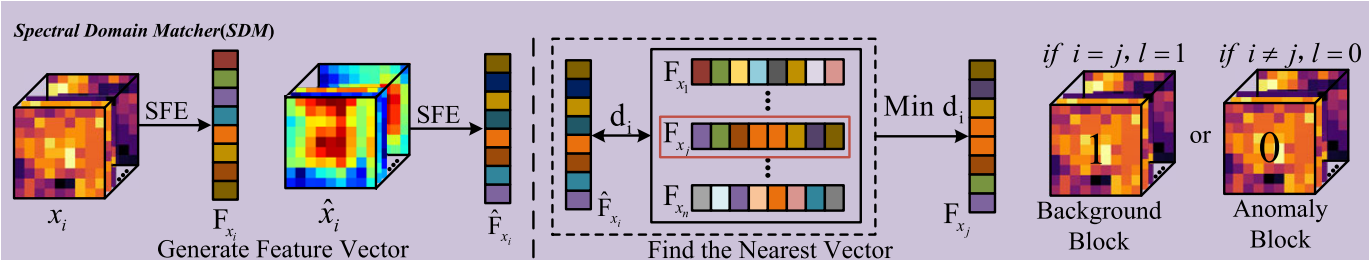


Fig. 3. Workflow of SDM.

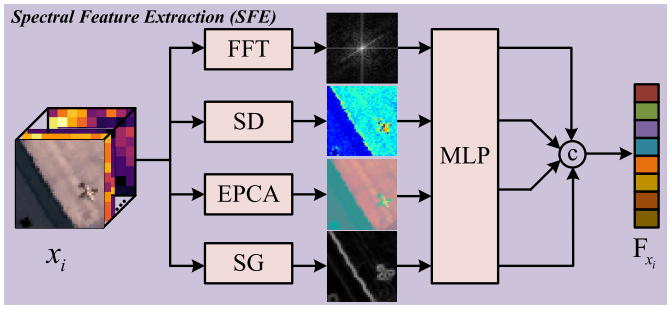


Fig. 4. SFE. Multidimensional features, including Fourier components (FFT), SD, eigenvalue PCA (EPCA), and SG, are extracted in parallel 1. These features are subsequently refined through an MLP and concatenated to generate the final discriminative feature vector $\mathbf{F}_{x_i}$.

and performing overlap averaging to ensure spatial consistency across the entire image.

B. Spectral-Domain Matcher

The SDM is designed to guide background reconstruction by providing preliminary classification labels through a similarity-driven matching mechanism. As shown in Fig. 3, it processes original patches x_i and their reconstructed versions $\hat{x}_i$ to generate matching labels $l_i \in \{0, 1\}$.

1) *Spectral Feature Extraction (SFE):*
The SDM leverages FFT as its primary technique, combined with spectral derivatives (SD), (PCA, and spatial gradients (SG). Specifically, FFT is applied to capture spatial texture patterns, while other descriptors extract material properties. These are fused to generate robust spatial–spectral feature vectors $\mathbf{F}_{x_i} \in \mathbb{R}^n$. As illustrated in Fig. 4, the SFE unit extracts multidimensional features through four parallel branches, which are refined by a multilayer perceptron (MLP) and concatenated to form the final feature vector $\mathbf{F}_{x_i}$.
2) *Background Template Initialization:*
To establish a benchmark for the background, the SDM initializes a background template $\mathbf{F}_{template}$ by aggregating the spectral features of all patches within the same scene. Specifically, the template is defined as the mean of the entire set of patch features $\{F_{x_i}\}_{i=1}^{N}$

$$\mathbf{F}_{template} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{F}_{x_i}. \tag{1}$$

This global average represents the dominant spectral characteristics of the scene, providing a stable reference for identifying anomalies.

3) *Nearest Vector Matching and Labeling:*
For a reconstructed patch, the module extracts its feature $\hat{F}_{x_i}$ and performs a search across the original feature library $\{F_{x_k}\}_{k=1}^{N}$ to find the nearest original vector $F_{x_j} = \arg\min \|\hat{F}_{x_i} - F\|_2$. If F_{x_j} aligns with the original background characteristics, it confirms successful reconstruction.
Simultaneously, the SDM computes the L_2 distance $d_i = \|\mathbf{F}_{x_i} - \mathbf{F}_{template}\|_2$ for each patch. To achieve adaptive separation, a dynamic threshold T is established

$$T = \text{Mean}\left(\{d_i\}_{i=1}^{N}\right) + 2 \cdot \text{Std}\left(\{d_i\}_{i=1}^{N}\right). \tag{2}$$

The matching label $\mathbf{l}_i$ is determined based on the feature distance and a scene consistency check. To ensure the comparison remains valid within the specific data distribution, we define $\mathcal{S}(\cdot)$ as a scene identifier function. The label is generated as follows:

$$l_i = \begin{cases} 1, & \text{if } d_i < T \text{ and } \mathcal{S}\left(\mathbf{F}_{\mathbf{x}_i}\right) = \mathcal{S}\left(\mathbf{F}_{template}\right) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where the condition $\mathcal{S}(\mathbf{F}_{\mathbf{x}_i}) = \mathcal{S}(\mathbf{F}_{\text{template}})$ ensures that the patch $\mathbf{x}_i$ is strictly compared against the background statistics of its own source image. This mechanism preliminarily classifies patches as background ($\mathbf{l}_i = 1$) or anomaly ($\mathbf{l}_i = 0$).
4) *Dynamic Update Mechanism:*
To ensure the matching remains valid, the SDM employs a staged update strategy. In the early stage (first 50 iterations), the template and labels are updated every five iterations to adapt to initial reconstruction errors. In the later stage, the frequency is reduced to every 25 iterations to stabilize the classification.
5) *Integration With QSME-Net:*
The outputs of the SDM, specifically the feature vector $\mathbf{F}_{x_i}$ and the matching label l_i, are directly fed into the QSCE module. Crucially, the SDM employs a joint spatial–spectral frequency analysis strategy, where spatial frequency components capture texture anomalies while the spectral frequency analysis is fundamental for ensuring spectral continuity within the QSCE's gating mechanism. Specifically, since raw spectral signatures often contain high-frequency noise and band-specific fluctuations, the SDM transforms these signatures into the frequency domain and retains the dominant low-to-mid frequency components to extract the intrinsic spectral envelope of the material. This process effectively filters out high-frequency discontinuities caused

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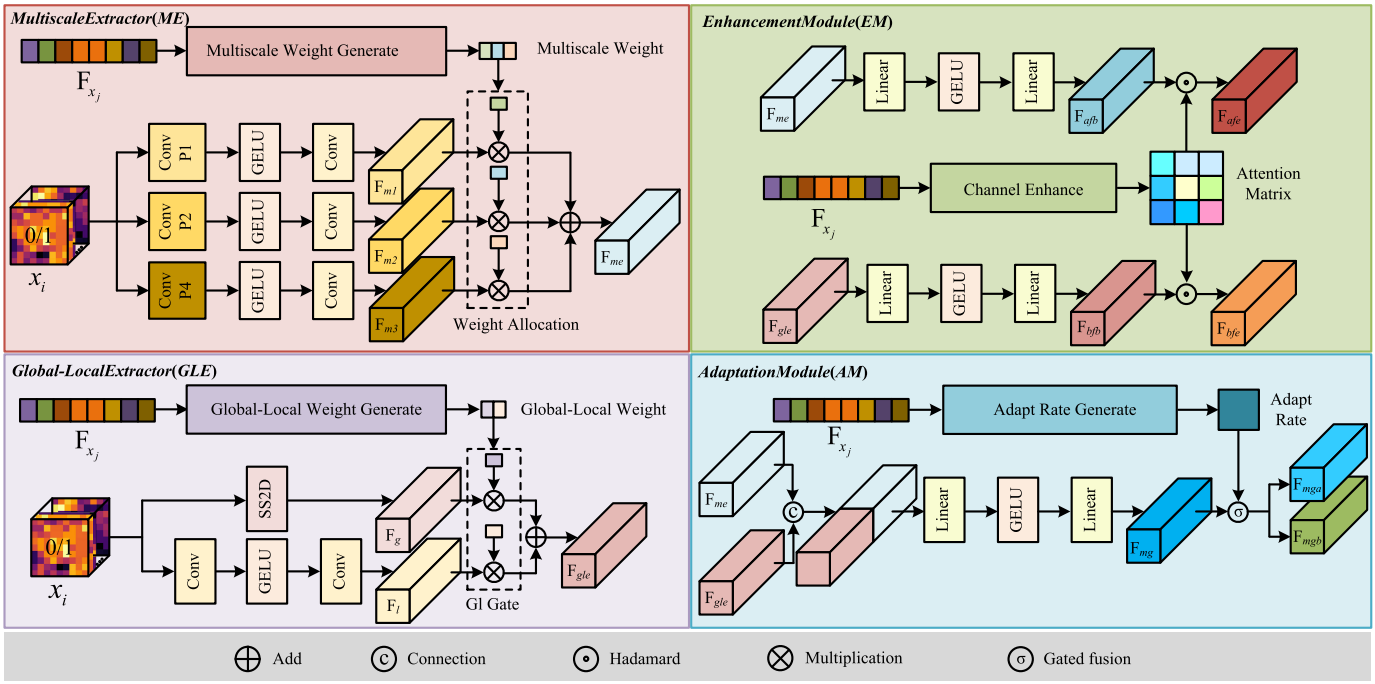


Fig. 5. Components of QSCE are as follows: ME, Global–LocalExtractor, EnhancementModule, and AdaptationModule.

by sensor noise, ensuring that the gating weights in the QSCE, derived from this stabilized spectral representation, exhibit inherent smoothness and coherence across the spectral dimension. Consequently, the dynamic fusion process is guided by robust material properties rather than volatile noise artifacts.

C. Quad-State Cross Enhancer

The QSCE is a pivotal module in the QSME-Net framework for HSI anomaly detection. It aims to enhance feature representations by generating and fusing four state features: MultiscaleExtractor (ME) anomaly in Figs. 2 and 5, Global-LocalExtractor (GLE) background, ME background, and GLE anomaly. The QSCE leverages spectral features and matching labels from the SDM, utilizing the SS2D module of Mamba for global–local feature modeling. Through spectral-guided gating and dynamic fusion, it optimizes feature discriminability. This section details the QSCE's inputs and outputs, dual-branch structure, four-state feature generation, dynamic fusion, pseudocode, and integration with QSME-Net.

1) Inputs and Outputs: The QSCE accepts the following parameterized inputs.

a) *Patch Features:* $\mathbf{X} \in \mathbb{R}^{B\times k\times k\times D}$, mapped from input patches $\mathbf{B}_i \in \mathbb{R}^{C\times k\times k}$ by a convolutional head.
b) *Matched Spectral Features:* $\mathbf{F}_{x_j} \in \mathbb{R}^{B\times n}$, the n-dimensional spectral vectors provided by the SDM.
c) *Matching Labels:* $\mathbf{l} \in \{0,1\}^B$, indicating background (1) or anomaly (0).
d) *Enhanced Features:* $\mathbf{F}_{\text{out}} \in \mathbb{R}^{B\times k\times k\times D}$, the final refined output.

2) Dual-Branch Feature Extraction:

a) MultiscaleExtractor: The ME branch captures anomaly-specific spatial patterns using multiscale dilated convolutions with dilation rates $\{2^0, 2^1, 2^2\}$

$$\mathbf{F}^i_{\text{me}} = \text{Conv}_i(\mathbf{X}), \quad i \in \{1,2,3\}. \tag{4}$$

The features are fused via spectral-guided weights

$$\mathbf{W}_{\text{me}} = \text{Softmax}\left(\text{MLP}\left(\mathbf{F}_{x_j}\right)\right) \in \mathbb{R}^{B\times 3\times 1\times 1} \tag{5}$$

$$\mathbf{F}_{\text{me}} = \sum_{i=1}^{3} \mathbf{W}^i_{\text{me}} \cdot \mathbf{F}^i_{\text{me}} \tag{6}$$

b) GlobalLocalExtractor: The GLE branch combines long-range SS2D features and local texture features

$$\mathbf{F}_{\text{SS2D}} = \text{SS2D}(\mathbf{X}) \tag{7}$$

$$\mathbf{F}_{\text{Conv}} = \text{Conv}_{3\times 3}\left(\text{Conv}_{1\times 1}(\mathbf{X})\right). \tag{8}$$

The global–local gating weights $\mathbf{W}_{\text{gle}}$ are defined as follows:

$$\mathbf{W}_{\text{gle}} = \text{Softmax}\left(\text{MLP}\left(\mathbf{F}_{x_j}\right)\right) \in \mathbb{R}^{B\times 2\times 1\times 1}. \tag{9}$$

The fusion is performed as follows:

$$\mathbf{F}_{\text{gle}} = \mathbf{W}^1_{\text{gle}} \cdot \mathbf{F}_{\text{SS2D}} + \mathbf{W}^2_{\text{gle}} \cdot \mathbf{F}_{\text{Conv}}. \tag{10}$$

3) Four-State Feature Generation:

a) Enhancement module: The EM utilizes spectral-guided channel attention $\mathbf{A}_{\text{channel}}$ to amplify primary representations

$$\mathbf{A}_{\text{channel}} = \text{Sigmoid}\left(\text{MLP}\left(\mathbf{F}_{x_j}\right)\right) \in \mathbb{R}^{B\times D\times 1\times 1} \tag{11}$$

$$\mathbf{F}_{afe} = \mathbf{F}_{\text{me}} \cdot \mathbf{A}_{\text{channel}} + \text{MLP}(\mathbf{F}_{\text{me}}) \tag{12}$$

$$\mathbf{F}_{bfe} = \mathbf{F}_{\text{gle}} \cdot \mathbf{A}_{\text{channel}} + \text{MLP}\left(\mathbf{F}_{\text{gle}}\right) \tag{13}$$

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b) Adaptation module: The AM generates fallback features $\mathbf{F}_{mga}$ and $\mathbf{F}_{mgb}$ to handle matching errors, using an adaptive gate $\mathbf{G}$

$$\mathbf{G} = \text{Sigmoid}\left(\text{MLP}\left(\mathbf{F}_{x_j}\right)\right) \in \mathbb{R}^{B\times 1\times 1\times 1} \tag{14}$$

$$\mathbf{F}_{mga} = \mathbf{G}\cdot\text{MLP}\left(\left[\mathbf{F}_{\text{me}}, \mathbf{F}_{\text{gle}}\right]\right) + (1-\mathbf{G})\cdot\mathbf{F}_{\text{me}} \tag{15}$$

$$\mathbf{F}_{mgb} = \mathbf{G}\cdot\text{MLP}\left(\left[\mathbf{F}_{\text{gle}}, \mathbf{F}_{\text{me}}\right]\right) + (1-\mathbf{G})\cdot\mathbf{F}_{\text{gle}}. \tag{16}$$

4) Dynamic Fusion and Refinement: The states are dynamically fused based on labels $\mathbf{l}$ and fallback weight λ

$$\begin{aligned}\mathbf{F}_{\text{out}} = {} & (1-\mathbf{l})\cdot\mathbf{F}_{afe} + \mathbf{l}\cdot\mathbf{F}_{bfe} \\ & + \lambda\cdot\mathbf{l}\cdot\mathbf{F}_{mga} + \lambda\cdot(1-\mathbf{l})\cdot\mathbf{F}_{mgb}.\end{aligned} \tag{17}$$

The final output is refined via an MLP and a 3×3 convolution

$$\mathbf{F}_{\text{out}} = \text{Conv}_{3\times 3}\left(\text{MLP}\left(\mathbf{F}_{\text{out}}\right)\right). \tag{18}$$

Algorithm 1 Quad-State Cross Enhancer

Require: Patch features $\mathbf{X}\in\mathbb{R}^{B\times k\times k\times D}$, Matched spectral features $\mathbf{F}_{x_j}\in\mathbb{R}^{B\times n}$, Matching labels $\mathbf{l}\in\{0,1\}^B$
Ensure: Enhanced features $\mathbf{F}_{\text{out}}\in\mathbb{R}^{B\times k\times k\times D}$

// Step 1: Multiscale Feature Extraction (ME)
$\mathbf{F}^1_{\text{me}} \leftarrow \text{Conv}_{r=1}(\mathbf{X})$
$\mathbf{F}^2_{\text{me}} \leftarrow \text{Conv}_{r=2}(\mathbf{X})$
$\mathbf{F}^3_{\text{me}} \leftarrow \text{Conv}_{r=4}(\mathbf{X})$
$\mathbf{W}_{\text{me}} \leftarrow \text{Softmax}(\text{MLP}(\mathbf{F}_{x_j}))$
$\mathbf{F}_{\text{me}} \leftarrow \mathbf{W}^1_{\text{me}}\cdot\mathbf{F}^1_{\text{me}} + \mathbf{W}^2_{\text{me}}\cdot\mathbf{F}^2_{\text{me}} + \mathbf{W}^3_{\text{me}}\cdot\mathbf{F}^3_{\text{me}}$
// Step 2: Global-Local Feature Extraction (GLE)
$\mathbf{F}_{\text{SS2D}} \leftarrow \text{SS2D}(\mathbf{X})$
$\mathbf{F}_{\text{Conv}} \leftarrow \text{Conv}_{3\times 3}(\text{Conv}_{1\times 1}(\mathbf{X}))$
$\mathbf{W}_{\text{gle}} \leftarrow \text{Softmax}(\text{MLP}(\mathbf{F}_{x_j}))$
$\mathbf{F}_{\text{gle}} \leftarrow \mathbf{W}^1_{\text{gle}}\cdot\mathbf{F}_{\text{SS2D}} + \mathbf{W}^2_{\text{gle}}\cdot\mathbf{F}_{\text{Conv}}$
// Step 3: Four-State Generation (EM & AM)
$\mathbf{A}_{\text{channel}} \leftarrow \text{Sigmoid}(\text{MLP}(\mathbf{F}_{x_j}))$
$\mathbf{F}_{afe} \leftarrow \mathbf{F}_{\text{me}}\cdot\mathbf{A}_{\text{channel}} + \text{MLP}(\mathbf{F}_{\text{me}})$
$\mathbf{F}_{bfe} \leftarrow \mathbf{F}_{\text{gle}}\cdot\mathbf{A}_{\text{channel}} + \text{MLP}(\mathbf{F}_{\text{gle}})$
$\mathbf{F}_{\text{cat}} \leftarrow [\mathbf{F}_{\text{me}}, \mathbf{F}_{\text{gle}}]$
$\mathbf{G} \leftarrow \text{Sigmoid}(\text{MLP}(\mathbf{F}_{x_j}))$
$\mathbf{F}_{mga} \leftarrow \mathbf{G}\cdot\text{MLP}(\mathbf{F}_{\text{cat}}) + (1-\mathbf{G})\cdot\mathbf{F}_{\text{me}}$
$\mathbf{F}_{mgb} \leftarrow \mathbf{G}\cdot\text{MLP}(\mathbf{F}_{\text{cat}}) + (1-\mathbf{G})\cdot\mathbf{F}_{\text{gle}}$
// Step 4: Dynamic Fusion and Refinement
$\mathbf{F}_{\text{out}} \leftarrow (1-\mathbf{l})\cdot\mathbf{F}_{afe} + \mathbf{l}\cdot\mathbf{F}_{bfe} + \lambda\cdot\mathbf{l}\cdot\mathbf{F}_{mga} + \lambda\cdot(1-\mathbf{l})\cdot\mathbf{F}_{mgb}$
$\mathbf{F}_{\text{out}} \leftarrow \text{Conv}_{3\times 3}(\text{MLP}(\mathbf{F}_{\text{out}}))$
return $\mathbf{F}_{\text{out}}$

5) Pseudocode: The comprehensive logic of the QSCE is summarized in Algorithm 1.

6) Integration With QSME-Net: The QSCE's inputs $\mathbf{F}_{x_j}$ and $\mathbf{l}$ are sourced from the SDM, providing a robust spectral foundation for feature enhancement. By utilizing Fourier-based features to preserve spectral continuity, the spectral-guided gating in QSCE ensures that decisions are anchored to the inherent physical properties of the HSI data. The SS2D module within GLE facilitates consistent global–local modeling, while the four-state dynamic fusion ensures high detection accuracy by effectively managing matching label uncertainty.

Algorithm 2 Overall Workflow of QSME-Net

Part 1: Training Phase
Require: Hyperspectral Image I, Epochs E, Learning Rate η.
Ensure: Trained QSME-Net model Θ, final Background Template $F_{template}$.

Initialize QSME-Net model parameters Θ and Optimizer.
▷Initial SDM Run (Builds initial template)
Patches $\{x_i\} \leftarrow$ Patchify(I)
$\{F_{x_i}\}, F_{template} \leftarrow$ Algorithm 1($\{x_i\}$)
for $e = 1$ **to** E **do**
▷QSCE (Alg. 2) logic is applied internally
$x_{out} \leftarrow \Theta(\{x_i\}, F_{template})$ ▷Process all patches (in batches)
$L_{recon} = \text{Loss}(\{x_i\}, x_{out})$
Update Θ using ∇L_{recon} ▷Backward pass
if e % update_freq == 0 **then**
▷Dynamic SDM Template Update (Sec III-B-IV)
Update $F_{template}$ using Θ and all $\{x_i\}$.
end if
end for
return Θ, $F_{template}$

Part 2: Inference (Testing) Phase
Require: Test HSI I_{test}, trained model Θ, $F_{template}$.
Ensure: Final Anomaly Map M_{final}.

Patches $\{x_i\} \leftarrow$ Patchify(I_{test})
Initialize empty score list S_{list}
for each patch x_i **do**
$x_{out,i} \leftarrow \Theta(x_i, F_{template})$ ▷Forward pass, no grad
$S_i = \|x_i - x_{out,i}\|_2^2$
Append S_i to S_{list}
end for
$M_{final} \leftarrow$ Fold(S_{list}, I_{test}.shape) ▷Reconstruct map with overlap averaging
return Normalize(M_{final})

D. Mamba-Attention Fusion Framework

The MAFF is a core component of the encoder–decoder network, designed to progressively refine feature representations. As depicted in Fig. 2, the MAFF organizes its five submodules—SS2D, efficient channel attention (ECA), local difference convolution (LDC), attention, and bidirectional attention (BiAttn)—in a serial cascade. This serial arrangement allows for a sophisticated and multistage feature refinement process, where each module processes the output of the preceding one. A key architectural feature of MAFF is the synergistic integration of Mamba's selective scanning (SS2D) and a Transformer-based global attention (attention) mechanism. This synergy is designed to capture complementary aspects of the data. The SS2D module efficiently models long-range spatial–spectral dependencies by scanning the feature maps with linear complexity. Subsequently, the attention module models

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global contextual interactions by calculating "all-to-all" feature relationships. The synergy arises from their serial application: SS2D first efficiently embeds long-range context into the feature representation. The "Attention" module then operates on these context-aware features, allowing it to focus its comprehensive "all-to-all" computation on modeling the more complex, nonlocal interactions between features that are already globally informed. This two-stage process allows MAFF to achieve a deep and robust feature representation while balancing computational efficiency and modeling power.

1) Selective Scan 2-D: The SS2D module plays a crucial role in capturing long-range dependencies across spectral and spatial dimensions through a selective scanning mechanism. By efficiently aggregating global contextual information via state-space modeling, this module enhances context-awareness, making it particularly effective for detecting subtle anomalies dispersed throughout the hyperspectral data.

2) Local Difference Convolution: The LDC module emphasizes localized variations through learnable convolutional masks that are dynamically adjusted during training. These masks significantly improve the module's capability to identify fine-grained spatial discrepancies, essential for distinguishing anomalies from complex backgrounds.

3) Efficient Channel Attention: The ECA module implements a streamlined channelwise attention mechanism. Utilizing adaptive average pooling and compact 1-D convolutions, ECA dynamically enhances the spectral channels that carry the most discriminative information, thereby increasing the model's effectiveness in identifying spectral anomalies.

4) Attention: The Attention module leverages a transformer-based global attention mechanism to refine feature representations by dynamically weighting the importance of spatial and spectral features. Complementary to SS2D's selective scanning, this module enhances the model's ability to focus on globally significant patterns, thereby improving the detection of anomalies that span multiple regions or bands.

5) Bidirectional Attention: The BiAttn module integrates bidirectional spectral attention to effectively blend global and local spectral context. This adaptive attention mechanism dynamically modulates the spectral feature maps by utilizing global spectral statistics, thus refining the feature representation for more robust anomaly detection.

By integrating these specialized submodules, MAFF robustly enhances spectral–spatial feature extraction capabilities, directly supporting the QSCE module's effectiveness in managing complex anomaly and background scenarios.

E. Inference and Anomaly Map Generation

Once the QSME-Net has been trained, the complete model Θ and the final, stable background template $F_{template}$ are fixed. The inference (testing) phase for a new, unseen HSI I_{test} proceeds as follows.

1) Patching: The test image I_{test} is decomposed into overlapping patches $\{x_i\}$ using the same 9×9 window and stride of 3 employed during training. The spatial coordinates (r, c) for each patch's center pixel are recorded.

2) Forward Pass: Each patch x_i is passed through the trained network. Crucially, all model parameters (QSCE, Mamba–encoder–decoder) are frozen, and the SDM module operates in inference mode (i.e., it uses the fixed $F_{template}$ for matching, but no dynamic updates occur). The network performs the full forward pass

$$x_{out,i} = \Theta\left(x_i, F_{template}\right). \tag{19}$$

3) Score Calculation: The anomaly score S_i for each patch x_i is computed as the mean squared error (mse) or L_2-norm between the input patch and its reconstruction

$$S_i = \left|\left|x_i - x_{out,i}\right|\right|_2^2. \tag{20}$$

4) Map Aggregation: An empty anomaly map M and a counting map C (both initialized to 0 with the same spatial dimensions as I_{test}) are created. As the patches overlap, a single pixel at position (r, c) will receive scores from multiple patches. We aggregate these scores by adding the score S_i of each patch to all pixel locations it covers in M, and incrementing the corresponding locations in C.

5) Averaging and Normalization: The final anomaly map is generated by performing an elementwise division of the aggregated score map M by the count map C, which effectively averages the scores for overlapping pixels. This map is then normalized to a [0, 1] range for visualization

$$M_{final} = \text{Normalize}\left(M \odot C^{-1}\right). \tag{21}$$

IV. Experiments

The performance of QSME-Net was benchmarked against nine competing HAD techniques across six real HSI datasets. The results confirm that our proposed method substantially outperforms these existing approaches.

A. Datasets

1) AIRPORT-1 Dataset: This dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Los Angeles area. The image has a spatial size of 100×100 pixels, a spatial resolution of 7.1 m/pixel, and includes 205 spectral bands. The anomalies in this scene consist of several airplanes, which account for 144 abnormal pixels (1.44% of the total).

2) AIRPORT-2 Dataset: Also collected by the AVIRIS sensor over Los Angeles, this dataset has a spatial dimension of 100×100, a resolution of 7.1 m/pixel, and contains 205 spectral bands. In this image, two airplanes constitute the 87 anomalous pixels that need to be detected, making up 0.87% of the total pixels.

3) AIRPORT-4 Dataset: The dataset was acquired over Gulfport using the AVIRIS sensor and features a spatial resolution of 3.4 m/pixel. In this study, we utilized the standard preprocessed version of the dataset commonly used in anomaly detection benchmarks. In this version, bands affected by water vapor absorption and low signal-to-noise ratios have been pre-excluded, resulting in a high-quality image with 191 spectral bands and a spatial size of 100×100. The anomalous targets are three airplanes of various sizes, corresponding to 60 abnormal pixels (0.6% of the total).

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4) URBAN-1 Dataset: Captured by the AVIRIS sensor over the Texas Coast region, this dataset consists of a 100 × 100 pixel image with a spatial resolution of 17.2 m/pixel and 204 spectral bands. The anomalous targets in this dataset account for 0.67% of the total pixels.

5) BEACH-1 Dataset: This dataset, from the vicinity of Cat Island, has a spatial size of 150 × 150 pixels and a spatial resolution of 17.2 m/pixel. After noisy bands were excluded, 188 spectral bands were retained for the analysis. In this scene, a single airplane makes up the 19 anomalous pixels, which is 0.08% of the total.

6) BEACH-4 Dataset: The dataset was acquired over Pavia by the reflective optics system imaging spectrometer (ROSIS) sensor. It has a spatial resolution of 1.3 m/pixel, a size of 150 × 150, and contains 102 spectral bands after the removal of unsuitable ones. The primary anomalies are several vehicles on a bridge, which comprise 68 abnormal pixels (0.3% of the total).

B. Implementation Details

1) Comparison Methods: We benchmark QSME-Net against nine methods categorized into statistics-based (RX [13], KIFD [50]), representation-based (CRD [34], GTVLRR [51], PTA [33], and PCA-TLRSR [52]), and deep-learning-based approaches (Auto-AD [10], NL2Net [53], and BockNet [54]). These baselines were selected to cover the spectrum from classical statistical models and traditional optimization theories to state-of-the-art deep learning techniques, ensuring a comprehensive performance evaluation. All of the aforementioned methods have gained widespread attention and demonstrated competitive performance, with RX, in particular, being regarded as the most classic and foundational benchmark method in the field.

2) Experimental Settings: All experiments covered in this article are based on the PyTorch framework. The deep-learning-based methods are implemented on a cluster of NVIDIA GPUs (GeForce RTX 4060) and Intel Core i7 processors. The software stack runs on the Linux system Ubuntu 22.04, and the traditional methods run on MATLAB 2022a. We used the Adam optimizer with a learning rate set to $1e^{-4}$ and $1e^{-5}$ to optimize the performance of the QSME-Net model efficiently. For the architectural parameters, we set the patch size $k = 9$ and the feature channel dimension $D = 64$. The SFE unit generates an $n = 15$ dimensional vector for each patch. In the QSCE module, the hidden dimension of all MLPs is set to $D_{hid} = 64$, and the fallback weight λ is empirically determined to be 0.5.

3) Evaluation Metrics: We used evaluation metrics recognized in the field of HAD to assess the performance of QSME-Net as well as other comparative methods. The evaluation metrics include anomaly detection map, box plot, 2-D receiver operating characteristic (ROC) curve, and area under the ROC curve (AUC). Among them, the anomaly detection map can visually represent the background suppression effect and anomaly detection. Through the color change, we can quickly identify the location and intensity of the abnormal pixels to evaluate the effectiveness of QSME-Net. The box plot, also called statistical separability analysis, is used to evaluate the separability between background and anomaly. ROC presents the probability of detection (PD) versus the FAR for all possible thresholds, and curves close to the upper left corner indicate better detection performance. The AUC is determined by integrating the area beneath the ROC curve. A higher AUC value corresponds to better detection accuracy. An ideal AUC score is 1, representing a perfect scenario, where all anomalies are identified with a zero FAR.

C. Comparative Analysis

This section presents a comprehensive analysis of the detection performance of the ten competing methods across the six datasets. The anomaly maps generated by different HAD methods across the six experimental datasets are presented in Fig. 6. In the AIRPORT-1 Dataset, RX and CRD generate background clutter density significantly higher than QSME-Net's clean background, with target response intensity markedly weaker than QSME-Net's high-brightness performance; KIFD produces detected targets with blurred boundaries and irregular shapes, while QSME-Net presents clear and sharp target contours; GTVLRR and PTA show obvious target shape distortion and incompleteness, whereas QSME-Net maintains target geometric features highly consistent with ground truth. In the AIRPORT-2 Dataset's small target detection scenario, traditional methods universally suffer from excessive target region dispersion, with KIFD's target continuity is far inferior to the compact and concentrated performance of QSME-Net, and the GTVLRR and PTA generate numerous rectangular false alarms in stark contrast to QSME-Net's precise single-point detection; PCA-TLRSR exhibits significant stripe-like and block-like interference patterns across multiple datasets, severely affecting visual quality, while QSME-Net achieves smooth and uniform background suppression; Auto-AD method's overall detection response is generally dim and weak, with target prominence far below QSME-Net's intense highlighting performance. Regarding the deep learning-based competitors NL2Net and BockNet, although they demonstrate improved background suppression capabilities compared with traditional statistical models, they still exhibit noticeable limitations in target enhancement. As observed in the visualization results, NL2Net fails to completely eliminate structural background noise in complex scenes (e.g., in the AIRPORT-4 and URBAN-1 Datasets), resulting in residual clutter. BockNet, on the other hand, tends to suffer from target signal attenuation; its detected targets appear visually dim and lack the high-contrast saturation observed in QSME-Net, particularly in the BEACH-4 Dataset, where targets are barely visible. In contrast, QSME-Net consistently generates "red-hot" high-intensity target responses against a pristine dark background. Further analysis of the AIRPORT-4 Dataset, URBAN-1 Dataset, BEACH-1 Dataset, and BEACH-4 Dataset reveals that QSME-Net's target–background contrast is significantly superior to all traditional methods, with markedly lower background noise levels; particularly in the challenging environments of BEACH-1 Dataset and BEACH-4 Dataset, traditional methods universally experience degraded detection performance, weakened target responses, and increased background interference, while QSME-Net maintains stable

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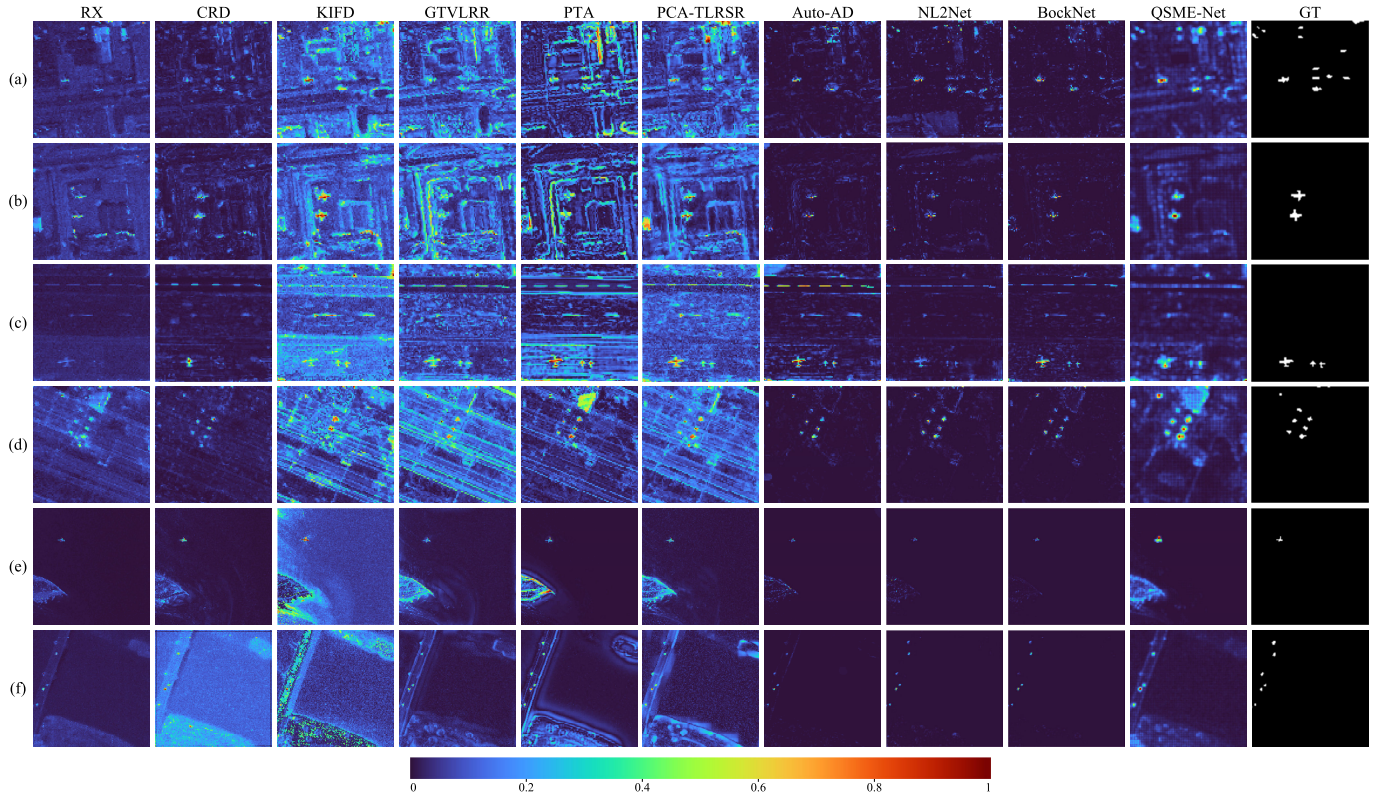


Fig. 6. Anomaly maps generated by different HAD methods across the six experimental datasets. (a) AIRPORT-1, (b) AIRPORT-2, (c) AIRPORT-4, (d) URBAN-1, (e) BEACH-1, and (f) BEACH-4.

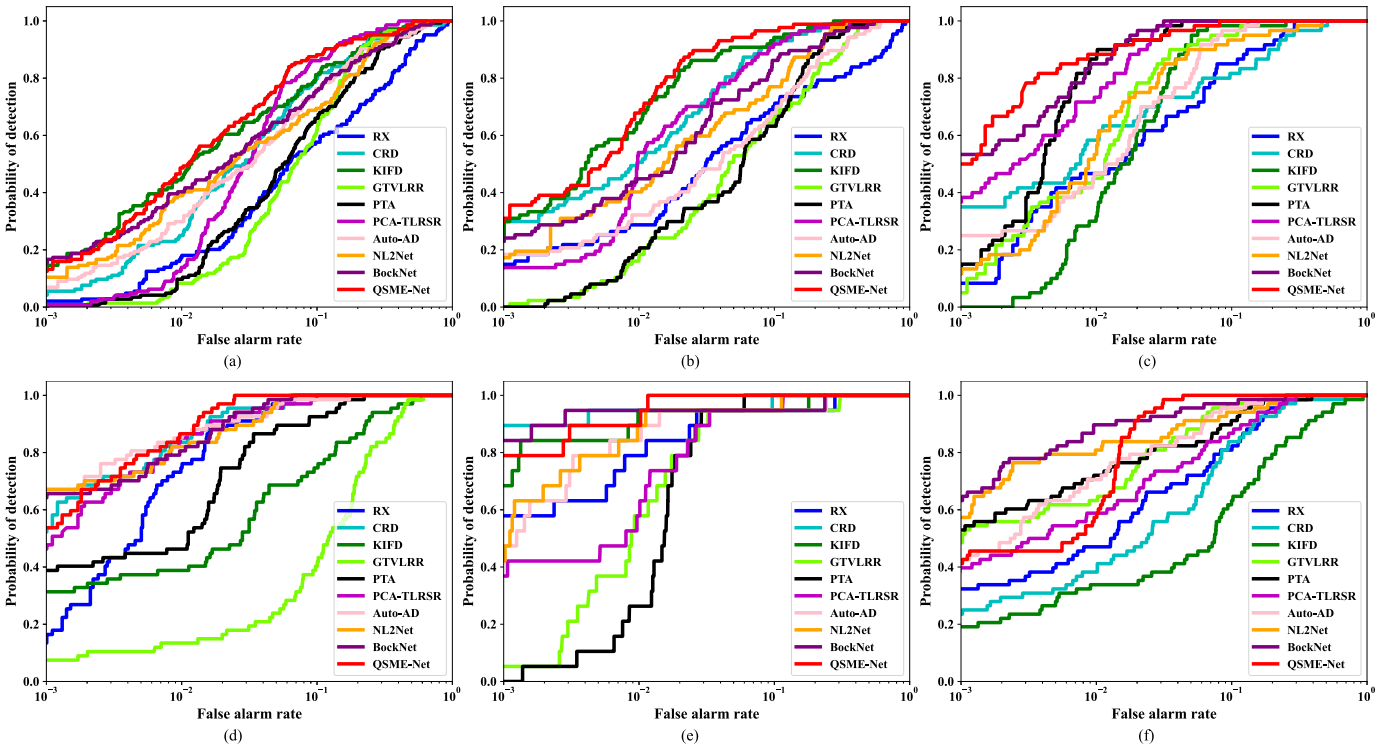


Fig. 7. Comparison of detection performance using 2-D ROC curves. Each plot shows the results for all methods on one of the six datasets. (a) AIRPORT-1, (b) AIRPORT-2, (c) AIRPORT-4, (d) URBAN-1, (e) BEACH-1, and (f) BEACH-4.

high-quality detection results with accurate target localization and clear boundaries. The results of heat map visualization further confirm that QSME-Net achieves optimal target response intensity and best background suppression effects across all tested datasets, demonstrating significant technical advantages over existing methods in terms of detection accuracy, noise

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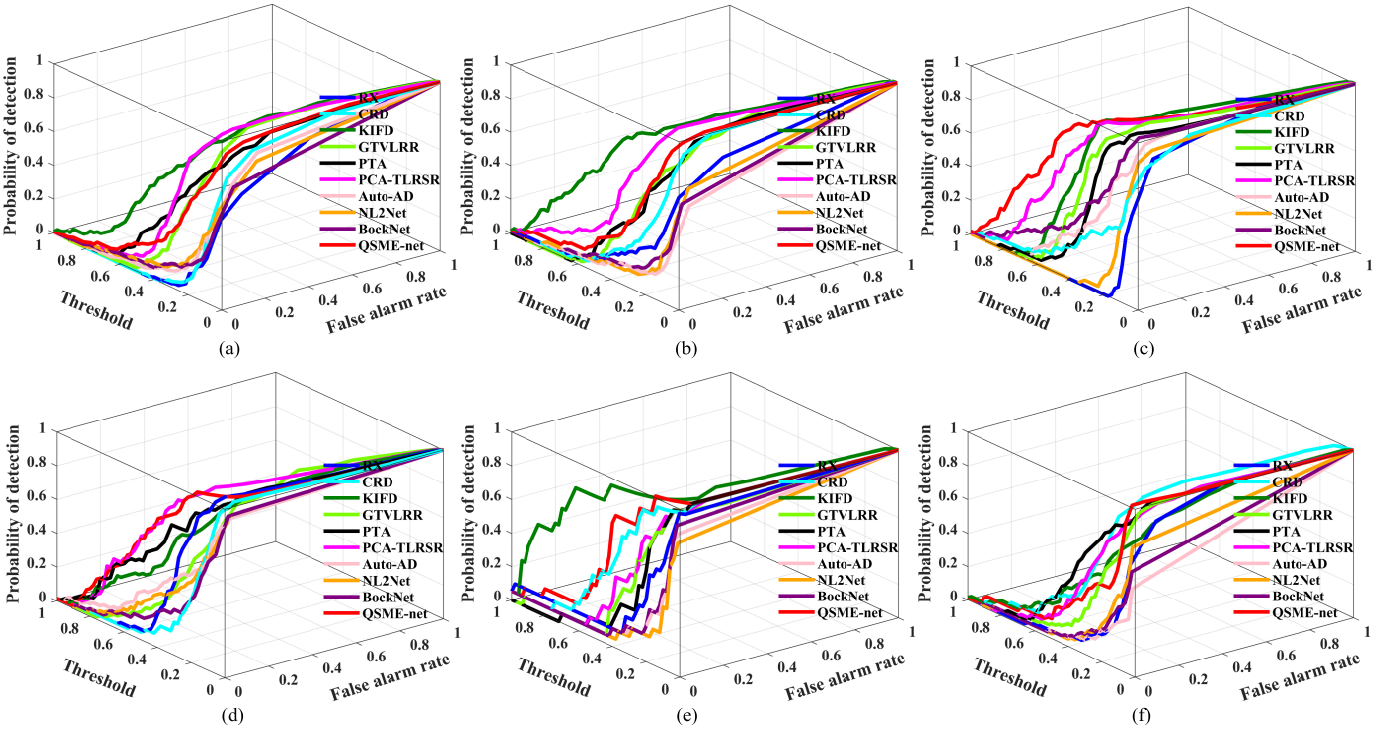


Fig. 8. Comparison of detection performance using 3-D ROC curves. Each plot shows the results for all methods on one of the six datasets. (a) AIRPORT-1, (b) AIRPORT-2, (c) AIRPORT-4, (d) URBAN-1, (e) BEACH-1, and (f) BEACH-4.

suppression capability, target integrity preservation, and cross-scenario robustness.

The detection performance of the ten methods on the six datasets is visualized by the 2-D ROC curves in Fig. 7 and the 3-D ROC curves in Fig. 8. The 2-D ROC curve analysis demonstrates that QSME-Net exhibits significant performance advantages in all test datasets. Its ROC curves closely adhere to the upper left boundary, maintaining high detection probabilities even under extremely low-FAR conditions. Specifically, on the AIRPORT-1 dataset, when the FAR is 10^{-3}, the detection probability of QSME-Net approaches the maximum value, while traditional methods such as RX and CRD show significantly reduced detection probabilities with relatively gradual upward trends in their curves. The ROC curve of KIFD exhibits notable fluctuations in the moderate FAR region, reflecting inconsistent detection performance. Although GTVLRR and PTA achieve high detection probabilities in high-FAR regions, their performance deteriorates sharply in the practically relevant low-FAR range. In contrast, the PCA-TLRSR shows a relatively low and flat ROC curve position, while the Auto-AD method's ROC curve lies closest to the diagonal, indicating relatively poor detection performance. Regarding the deep learning-based comparisons, the performance varies significantly. The Auto-AD method, despite employing deep neural networks, fails to effectively capture target features in these scenarios, with its ROC curve lying closest to the diagonal, indicating relatively poor detection performance. Conversely, the more advanced deep learning competitors, NL2Net and BockNet, show much stronger capabilities, generally outperforming traditional algorithms and forming a competitive second tier. However, as observed across the datasets, their curves still exhibit a noticeable gap beneath the QSME-Net boundary, particularly in the critical low-FAR region (10^{-3} to 10^{-2}), where QSME-Net maintains a decisive lead. In the small target detection scenario of the AIRPORT-2 dataset, the performance of traditional methods further deteriorates with overall downward shifts and shape distortions in their ROC curves, while QSME-Net maintains the shape and position of the near-ideal curve. On the AIRPORT-4 and URBAN-1 datasets, QSME-Net's ROC curves exhibit steep upward trends in the low FAR region, rapidly reaching a high detection probability plateau; here, BockNet demonstrates relatively robust performance as the runner-up, whereas NL2Net shows a slower convergence rate compared to QSME-Net. On the most challenging BEACH-1 and BEACH-4 datasets, traditional methods' ROC curves generally show significantly lower right shifts, with some methods failing to achieve satisfactory detection probabilities even at relatively high FARs. In these complex texture scenarios, although BockNet and NL2Net maintain stable curve trajectories significantly above the statistical methods and Auto-AD, they still exhibit delayed saturation compared to the steep ascent of QSME-Net, which consistently maintains excellent curve characteristics. The 3-D ROC analysis provides richer performance details. QSME-Net forms highly consistent peak-shaped surfaces across all datasets, with stable peak positions and smooth surfaces, indicating its ability to maintain stable high performance under different threshold settings. In contrast, traditional methods' 3-D ROC surfaces exhibit obvious topographical variations and multiple local

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TABLE I
PERFORMANCE COMPARISON OF HAD METHODS BASED ON AUC VALUES

Dataset	RX	CRD	KIFD	GTVLRR	PTA	PCA-TLRSR	Auto-AD	NL2Net	BockNet	**QSME-Net**
AIRPORT-1	0.8221	0.9212	0.9396	0.9012	0.8809	0.9455	0.9275	0.9114	0.9156	**0.9507**
AIRPORT-2	0.8404	0.9681	0.9785	0.8687	0.9085	0.9664	0.9192	0.9423	0.9453	**0.9844**
AIRPORT-4	0.9526	0.9445	0.9760	0.9875	0.9946	0.9931	0.9848	0.9690	0.9956	**0.9958**
URBAN-1	0.9907	0.9940	0.9240	0.9520	0.9757	0.9925	0.9903	0.9929	0.9940	**0.9964**
BEACH-1	0.9807	0.9946	0.9893	0.9702	0.9831	0.9854	0.9855	0.9918	0.9872	**0.9983**
BEACH-4	0.9538	0.9407	0.8694	0.9754	0.9752	0.9636	0.9834	0.9821	0.9909	**0.9916**
Average	0.9234	0.9605	0.9461	0.9425	0.9530	0.9744	0.9651	0.9649	0.9714	**0.9862**

optima. The RX and CRD show relatively flat surfaces with generally low heights; KIFD and GTVLRR display obvious ridge-like protrusions and valley-like depressions, reflecting their sensitivity to parameter settings; PCA-TLRSR and Auto-AD generally show low surface heights with irregular shapes. Similarly, while the deep learning competitors NL2Net and BockNet achieve visibly higher detection surfaces compared to the statistical models, they fail to maintain the surface continuity observed in the proposed method. Their 3-D trajectories exhibit noticeable fluctuations and steeper gradients along the threshold axis—particularly evident in the fragmented surface structures on the AIRPORT-4 and BEACH-1 datasets—indicating that their detection performance is more sensitive to threshold selection and lacks the robustness of QSME-Net's expansive high-probability plateau. Quantitative analysis results further validate the above observations. As shown in Table I, QSME-Net achieves the highest AUC values across all six test datasets, with an average AUC of 0.9862. This performance represents a significant lead over the strongest statistical method, PCA-TLRSR (0.9744), and the top-performing deep learning competitor, BockNet (0.9714). QSME-Net's AUC values range from [0.9507, 0.9983] with a standard deviation of 0.0187, indicating good stability and generalization capability. Specifically comparing the performance gains, QSME-Net achieves an 8.88 percentage point average AUC improvement over the baseline RX detector. In the realm of deep learning comparisons, while BockNet and NL2Net have shown improvements over basic models, QSME-Net still maintains a distinct advantage; it surpasses BockNet, Auto-AD, and NL2Net by margins of 1.48%, 2.11%, and 2.13%, respectively. The comprehensive analysis demonstrates that QSME-Net not only outperforms comparison methods in single-point performance but, more importantly, exhibits excellent consistency and stability throughout the entire operational parameter space.

The box-whisker plots in Fig. 9 compare the statistical score distributions of the competing methods. The red boxes (anomalies) and blue boxes (background) allow for a visual assessment of performance. A greater distance between the boxes indicates better separability, and a smaller blue box indicates stronger background suppression. The plots reveal that QSME-Net is highly effective at both, outperforming the other methods. In the anomaly detection category, QSME-Net achieves detection statistics close to 1.0, with a tightly concentrated distribution, indicating its ability to efficiently and consistently identify anomalous regions with a low false positive rate. Simultaneously, its background detection statistics are close to 0.0, demonstrating that QSME-Net effectively suppresses background interference. In comparison, Auto-AD and PCA-TLRSR also perform well in anomaly detection, with statistics close to 1.0, but their background suppression is less effective, leading to occasional false positives in the background and some variability in performance. In comparison, Auto-AD and PCA-TLRSR perform well in anomaly detection, with statistics close to 1.0, but their background suppression is less effective, leading to occasional false positives in the background and some variability in performance. Conversely, the deep learning-based NL2Net and BockNet demonstrate superior background suppression capabilities, with background statistics tightly compressed nearly 0, comparable with QSME-Net. However, a distinct limitation is observed in their anomaly distributions: their red boxes are situated significantly lower on the vertical axis compared to QSME-Net, indicating that while they effectively silence the background, they fail to sufficiently amplify the target signal intensity. On the other hand, RX, CRD, and GTVLRR show significantly poorer performance, especially in anomaly detection, as their box plots are more widely spread, indicating that these methods exhibit high variability across different datasets and fail to effectively suppress background interference, resulting in frequent false alarms. Overall, QSME-Net consistently outperforms other methods across all six datasets, providing a reliable balance between accurate anomaly detection and effective background suppression, making it the most superior method for HAD.

D. Ablation Study

To evaluate the effectiveness of QSME-Net components in the HSI anomaly detection task, we conducted an ablation study comparing three configurations: the baseline model (Base), Base + SDM, and Base + SDM + QSCE. The results, presented in Table II, report AUC scores across six scenarios, with background regions marked by yellow dashed boxes and anomaly regions by red dashed boxes (see Fig. 10). Below,

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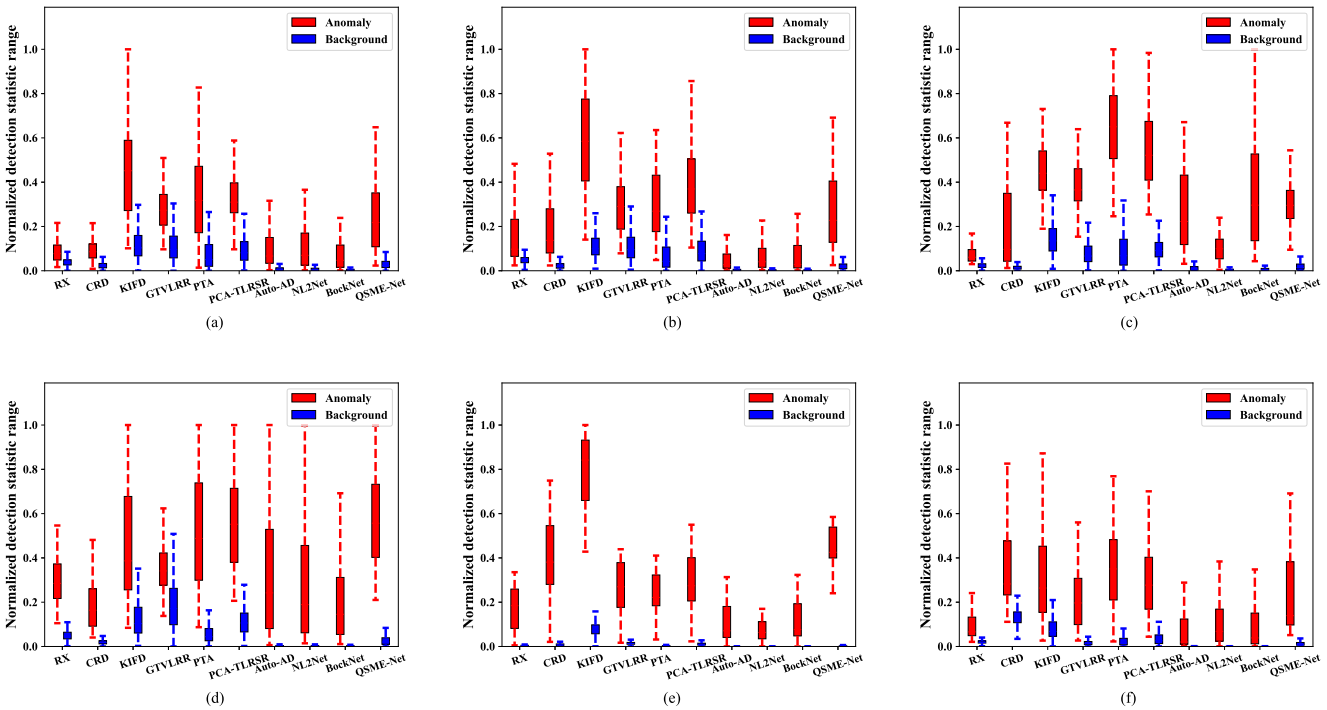


Fig. 9. Box plot analysis of method performance on the six datasets. (a) AIRPORT-1, (b) AIRPORT-2, (c) AIRPORT-4, (d) URBAN-1, (e) BEACH-1, and (f) BEACH-4.

TABLE II
ABLATION STUDY OF QSME-NET ON SIX DATASE TS

Models	AIRPORT-1	AIRPORT-2	AIRPORT-4	URBAN-1	BEACH-1	BEACH-4
Base	0.8655	0.9226	0.9797	0.9884	0.9725	0.8518
Base+SDM	0.9079	0.9743	0.9867	0.9907	0.9939	0.9844
Base+SDM+QSCE	**0.9507**	**0.9844**	**0.9958**	**0.9964**	**0.9983**	**0.9916**

TABLE III
SENSITIVITY ANALYSIS OF THE SECONDARY WEIGHT λ IN QSCE (AUC SCORE)

λ	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
AIRPORT-1	0.9372	0.9419	0.9469	0.9488	0.9498	**0.9507**	0.9498	0.9487	0.9462	0.9430	0.9384
AIRPORT-2	0.9734	0.9772	0.9815	0.9829	0.9836	**0.9844**	0.9836	0.9828	0.9808	0.9783	0.9746
AIRPORT-4	0.9873	0.9902	0.9938	0.9947	0.9951	**0.9958**	0.9951	0.9946	0.9931	0.9913	0.9885
URBAN-1	0.9881	0.9912	0.9936	0.9961	0.9953	**0.9964**	0.9962	0.9960	0.9940	0.9923	0.9882
BEACH-1	0.9912	0.9939	0.9959	0.9982	0.9973	**0.9983**	0.9981	0.9981	0.9964	0.9950	0.9914
BEACH-4	0.9770	0.9824	0.9865	0.9903	0.9903	**0.9916**	0.9911	0.9902	0.9870	0.9835	0.9772
Average	0.9757	0.9795	0.9830	0.9852	0.9852	**0.9862**	0.9857	0.9851	0.9829	0.9806	0.9764

we analyze the contributions of each component in enhancing the discrimination between background and anomaly regions based on their operational essence.

1) Baseline Model (Base): The Base model integrates global context modeling and local feature extraction to capture spatial and spectral characteristics of HSIs. Its core operation lies in balancing global and local information to generate initial anomaly detection residual maps. As shown in Table II, the Base model achieves robust baseline performance across all scenarios, with AUC scores of 0.9884 on URBAN-1 and 0.9725 on BEACH-1, indicating effective capture of background region (yellow dashed box) spectral consistency and preliminary identification of anomaly regions (red dashed box). However, in complex scenarios like BEACH-4 (0.8518), performance is notably lower, likely due to intricate background textures or spectral heterogeneity causing misdetection or omission of anomalies. Fig. 10(b) shows clear red dashed boxes in simpler scenarios but blurred anomaly boundaries and increased background misdetections in complex cases.

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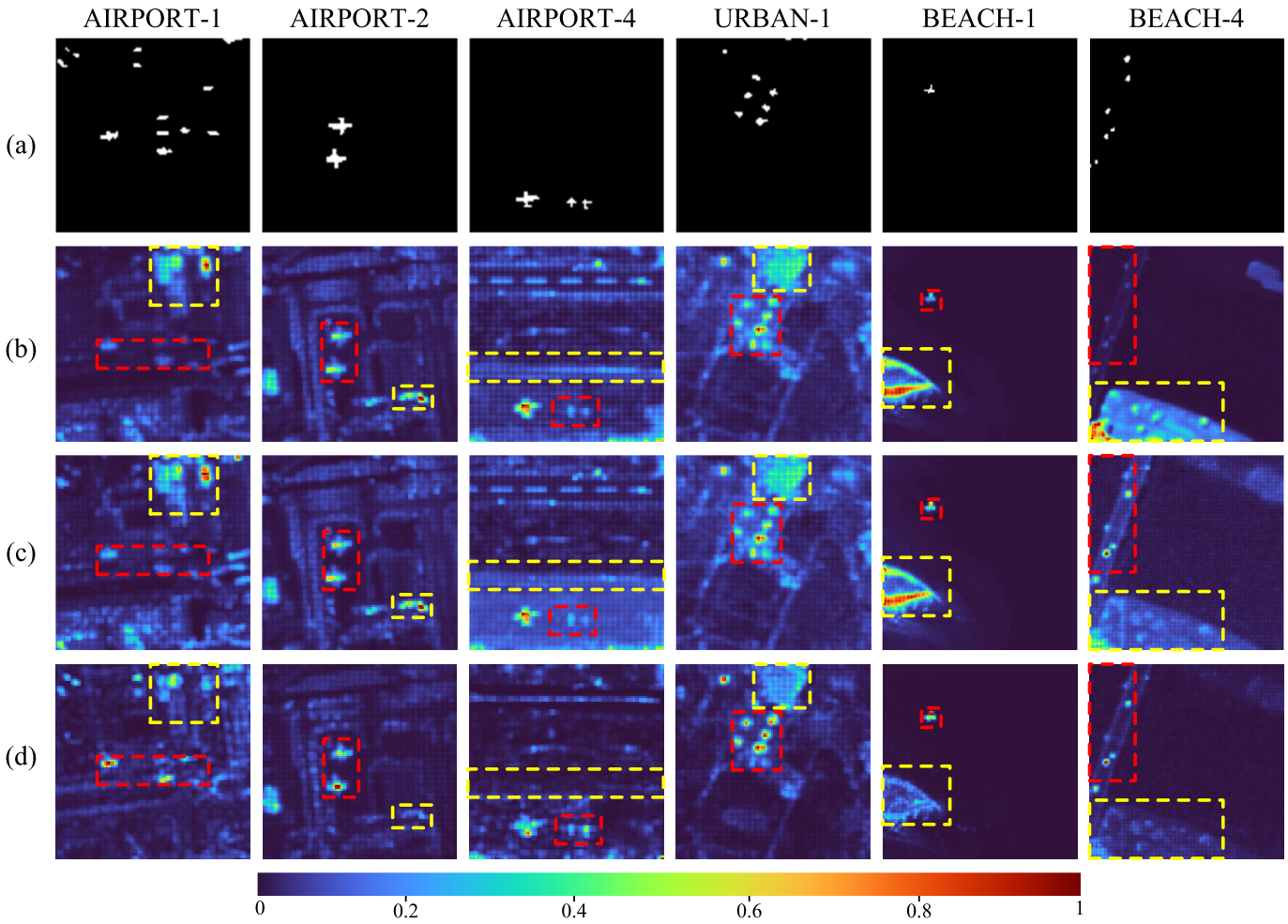


Fig. 10. Visual results of the ablation study. (a) Ground-truth maps, and the corresponding anomaly maps generated by (b) base model, (c) base model with SDM, and (d) full model. Yellow and red boxes highlight the background and anomaly regions, respectively, for comparison.

TABLE IV
ABLATION STUDY OF DIFFERENT SDM UPDATE SCHEDULES ON AIRPORT-1 AND BEACH-4 DATASETS

Update Schedule	AIRPORT-1		BEACH-4	
	AUC	Time (s)	AUC	Time (s)
Static (No update)	0.9215	**26.5**	0.9650	**55.2**
Frequent (Every 5)	0.9492	39.8	0.9905	82.4
Infrequent (Every 25)	0.9410	28.1	0.9820	58.5
Proposed (Staged)	**0.9507**	30.1	**0.9916**	61.3

TABLE V
ABLATION STUDY OF DILATION RATE COMBINATIONS IN ME

Rates	AIRPORT-1 (AUC)	BEACH-4 (AUC)
$\{1, 1, 1\}$	0.9385	0.9790
$\{1, 3, 5\}$	0.9462	0.9885
$\{1, 2, 4\}$	**0.9507**	**0.9916**

2) SDM Module (Base + SDM): The SDM module enhances the background region modeling through the frequency analysis along with spectral dimension feature extraction and similarity matching. Its core operation involves leveraging low-frequency spatial and spectral features to dynamically distinguish background from anomaly regions, improving feature representation robustness. Table II shows significant performance improvements with Base + SDM, with BEACH-1 increasing from 0.9725 to 0.9939 and BEACH-4 from 0.8518 to 0.9844 (a gain of 13.3%). The enhancement of SDM stems from capturing spatial consistency and spectral characteristics of background regions (yellow dashed boxes), substantially reducing confusion with anomaly regions (red dashed boxes), particularly in complex scenarios like BEACH-4. Fig. 10 (c) clearly shows sharper red dashed box boundaries and fewer background misdetections, highlighting the robustness of SDM.

3) QSCE Module (Base + SDM + QSCE): The QSCE module further optimizes anomaly detection through the multiscale feature extraction, global–local feature fusion, and adaptive enhancement. Its core operation uses spectrum-guided attention and dynamic feature adjustment to enhance the anomaly region saliency while suppressing the background interference. Table II indicates that Base + SDM + QSCE achieves the best performance across all scenarios, with AUC scores of 0.9983 on BEACH-1, 0.9916 on BEACH-4, and 0.9507 on AIRPORT-1 (a 9.8% improvement

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TABLE VI
RUNTIME (IN SECONDS) COMPARISON OF DIFFERENT ANOMALY DETECTION METHODS ON SIX HYPERSPECTRAL DATASETS

METHODS	AIRPORT-1	AIRPORT-2	AIRPORT-4	URBAN-1	BEACH-1	BEACH-4
RX	0.06	0.05	0.04	0.05	0.09	0.06
CRD	16.95	5.06	12.40	2.45	193.15	2.89
KIFD	33.99	33.99	32.78	32.56	37.70	43.51
GTVLRR	89.85	85.15	120.73	88.24	185.08	157.20
PTA	15.45	15.15	14.34	13.74	43.82	23.49
PCA-TLRSR	1.86	2.12	6.91	6.14	12.49	3.24
Auto-AD	9.74	10.78	6.31	8.82	18.49	22.13
NL2Net	122.09	123.24	122.11	121.25	207.74	202.51
BockNet	47.65	47.91	47.69	47.57	118.66	106.83
QSME-Net	30.07	30.10	30.24	29.94	63.75	61.27

over Base). QSCE's gains arise from multiscale extraction capturing varied anomaly scales, global–local fusion balancing background and foreground, and adaptive enhancement dynamically adjusting feature weights. These mechanisms significantly improve anomaly region (red dashed box) detection accuracy in complex scenarios like BEACH-4 and AIRPORT-1, while maintaining the clear background region (yellow dashed box) separation. Fig. 10(d) shows optimal detection, with precise anomaly boundaries and minimal background misdetections, underscoring QSCE's adaptive capability.

4) Synergistic Effects: The synergy of SDM and QSCE drives QSME-Net's superior performance. SDM provides the initial background-anomaly differentiation through frequency-domain matching, generating reliable guidance signals. QSCE builds on this by refining anomaly representations via multiscale extraction, cross-region fusion, and adaptive enhancement. This synergy ensures robustness in complex backgrounds, as evidenced by BEACH-4's improvement from 0.8518 (Base) to 0.9916 (Base + SDM + QSCE). The training process, with iterative optimization and periodic matching updates, ensures convergence to high accuracy. The progression across Fig. 10(a)–(d) clearly illustrates a transition from coarse to refined detection, with red dashed boxes becoming sharper and yellow dashed box interference significantly reduced.

E. Parameter Sensitivity Analysis

In this section, we evaluate the impact of two critical hyperparameters on the performance of QSME-Net: the secondary fusion weight λ in the QSCE module and the update frequency of the SDM template.

1) Sensitivity of Fusion Weight λ: As discussed in Section III-C, the secondary weight λ balances the reliability of the primary spectral matching with the robustness provided by cross-state features. Table III and Fig. 11 illustrates the AUC performance on six datasets as λ varies from 0.0 to 1.0. The results exhibit a consistent "inverted-U" trend, where performance peaks at $\lambda = 0.5$. Lower values ($\lambda < 0.3$) fail to provide sufficient fault tolerance, while higher values

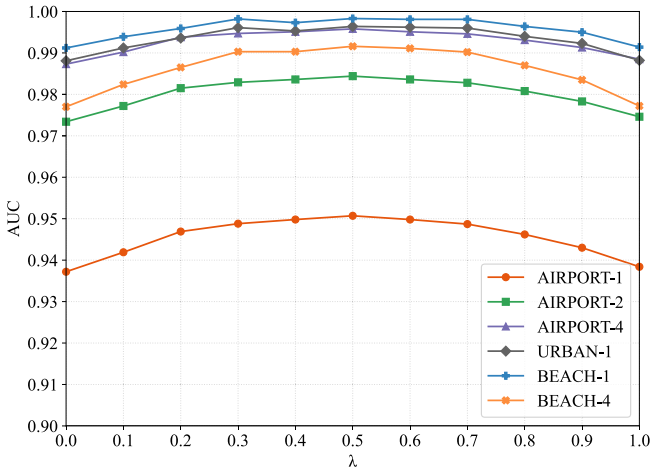


Fig. 11. Parameter λ sensitivity analysis.

($\lambda > 0.7$) introduce noise by over-fusing incorrect features. Consequently, we set $\lambda = 0.5$ as the default value.

2) Impact of SDM Update Schedule: To validate the optimality of our staged update strategy (updating every five iterations for the first 50 epochs and then every 25 iterations), we compared it against three fixed schedules: static (no update after initialization), frequent (every five epochs), and infrequent (every 25 epochs). As presented in Table IV, the comparative analysis on the AIRPORT-1 and BEACH-4 datasets reveals that while the *static* strategy yields the lowest accuracy and the *frequent* strategy incurs high computational costs, our proposed staged strategy achieves the highest AUC scores (0.9507 and 0.9916). Crucially, it maintains a runtime comparable to the *infrequent* schedule, thereby demonstrating an optimal tradeoff between early stage adaptability and late-stage computational efficiency.

3) Impact of Dilation Rates: To validate the architectural design of the multiscale extractor (ME), we analyzed the impact of different dilation rate combinations. We compared our proposed exponential setting $\{1, 2, 4\}$ against a fixed setting $\{1, 1, 1\}$ (representing standard convolutions without multiscale perception) and a sparse setting $\{1, 3, 5\}$. Table V summarizes the results on the AIRPORT-1 and BEACH-4 datasets. The

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TABLE VII
COMPUTATIONAL EFFICIENCY OF THREE DEEP LEARNING METHODS

AIRPORT-1	Auto-AD	NL2Net	QSME-Net
Params	3.25M	0.983M	0.58M
FLOPs	11.97G	7.972G	1.00G
Running Time	30.55s	122.09s	30.07s

fixed setting $\{1, 1, 1\}$ yields the lowest performance, demonstrating the necessity of enlarging the receptive field to capture the contextual information. While the sparse setting $\{1, 3, 5\}$ improves over the fixed one, it underperforms compared with the proposed method, likely due to the loss of local continuity (gridding effect). The proposed combination $\{1, 2, 4\}$ consistently achieves the highest AUC scores, confirming its effectiveness in capturing both fine-grained anomalies and broader background patterns.

F. Computational Efficiency

Table VI presents the runtime (in seconds) comparison of different anomaly detection methods on six hyperspectral datasets. Our efficiency analysis demonstrates that QSME-Net provides the most compact and computationally efficient architecture among the compared methods. We benchmarked QSME-Net against Auto-AD and NL2Net on the AIRPORT-1 dataset for a comprehensive comparison (see Table VII). Specifically, QSME-Net maintains a minimal parameter count of just 0.58M (significantly lower than NL2Net's 0.983M and Auto-AD's 3.25M) and requires only 1.00 GFLOPs of computation (compared with 7.972 GFLOPs for NL2Net and 11.97 GFLOPs for Auto-AD), making it exceptionally suitable for resource-constrained environments. Furthermore, QSME-Net exhibits superior execution speed with a runtime of 30.07 s, which is slightly faster than Auto-AD (30.55 s) and dramatically outperforms NL2Net's 122.09 s. Therefore, QSME-Net proves to be highly effective for real-time applications, achieving a distinct advantage in model size, computational cost, and processing speed simultaneously.

V. CONCLUSION

In this article, we introduce a new HAD method called QSME-Net, which guides the reconstruction of HSIs through a feature similarity-driven mechanism. It features two key components: the SDM for extracting spectral information and the QSCE for boosting the contrast between anomalies and the background using multiscale features. This work also demonstrates a new application of the Mamba architecture to HAD, providing a foundation for future research. Looking ahead, our goal is to improve the computational efficiency for real-time applications and to investigate the generalizability of our frequency-based hypothesis. This assumption, validated on high-frequency man-made anomalies, will require adaptive, domain-specific matching mechanisms to be extended to domains like agriculture and geology, where anomalies may be spatially diffuse and exhibit low-frequency characteristics.

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